

Contact

9407456834 (Mobile)
aravindai333@gmail.com

www.linkedin.com/in/
aravindb-9901151a3 (LinkedIn)
aravindbethapudi.github.io/
Portfolio/ (Portfolio)

Top Skills

Artificial Intelligence (AI)
Large Language Models (LLM)
Gen AI

Languages

Telugu
English

Certifications

Natural Language Processing with
Classification and Vector Spaces
LLM Engineer
Machine Learning with Python
Natural Language Processing with
Probabilistic Models
IBM Data Science Specialization

Aravind Bethapudi

AI Engineer | Generative AI • LangChain • RAG • Vertex AI • Agentic
AI • PyTorch • GCP | Building Scalable AI Systems
Denton, Texas, United States

Summary

I'm an AI & Machine Learning Engineer with 5+ years of experience designing and deploying Generative AI, LLM, and Data Science solutions across healthcare, industrial, and enterprise domains.

My work focuses on building end-to-end AI systems — from data engineering and model development to RAG pipelines, multi-agent architectures, and real-time AI applications using LangChain, LangGraph, PyTorch, TensorFlow, and Vertex AI.

I've deployed production-grade AI services using FastAPI, Docker, and Kubernetes across AWS, Azure, and GCP, with observability through MLflow, LangSmith, and Prometheus. I'm passionate about creating scalable, ethical, and explainable AI systems that bridge research and business impact.

Always exploring the next frontier in Agentic AI, automation, and LLM reasoning systems.

Experience

CVS Health

AI Engineer

October 2023 - Present (2 years 2 months)

Dallas, Texas, United States

- Designed and deployed RAG pipelines utilizing LangChain and GPT-4, achieving 94% accuracy in Q&A over 500K+ clinical documents.
- Automated claims triage processes with transformer models like BERT and RoBERTa, reducing manual review efforts by 35%.
- Implemented MLOps pipelines using SageMaker, MLflow, and Docker, decreasing deployment time from two weeks to just three days.
- Mentored ML engineers on CI/CD practices, model retraining, and production best practices to enhance team capabilities.

Honeywell

3 years 2 months

Data Scientist

July 2020 - July 2022 (2 years 1 month)

Bengaluru, Karnataka, India

- Developed predictive maintenance and forecasting models using TensorFlow and Azure ML, achieving annual savings of \$1M.
- Implemented real-time anomaly detection pipelines with autoencoders and Synapse, reducing incident response time by 60%.
- Designed Power BI dashboards to visualize KPIs, facilitating investment decisions exceeding \$50M.

Data Science Intern

June 2019 - June 2020 (1 year 1 month)

Bengaluru, Karnataka, India

- Developed a predictive efficiency model using XGBoost regression on over 5 GB of sensor telemetry, enhancing early inefficiency detection by 31%.
- Built an anomaly detection pipeline with Isolation Forest to identify unusual machine behavior prior to failures.
- Created interactive Power BI dashboards for real-time visualization of machine health and performance trends.
- Contributed to Honeywell's first predictive maintenance pilot, reducing unplanned downtime by approximately 15%.

Education

University of North Texas

Master's degree, Data Science · (August 2022 - May 2024)